WildChat Analytics Platform

Complete Tableau Dashboard Recreation Guide

Step-by-Step Instructions for Beginners

Group G-6 | WildChat Dataset Analysis

3 Dashboards · 23 Visualizations · 6 CSV Data Sources

Tableau Desktop 2023.1+

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1. Dashboard Overview

The WildChat Analytics Platform is a three-tab interactive dashboard built to analyze conversational AI usage patterns from the WildChat dataset. It provides operational, strategic, and safety insights across nearly 50,000 conversations.

1.1 Dashboard Tabs

The dashboard contains three main pages (tabs), each corresponding to a separate Tableau Dashboard sheet:

Tab Name	Icon	Purpose	Primary Data Sources
Executive Summary		High-level KPIs, trends, geography, model performance	conversations_clean.csv, daily_kpis.csv, geo_summary.csv
Operational Intelligence		Deep-dive on user behavior, segments, funnel, heatmaps	conversations_clean.csv, messages_sample.csv, prompt_categories.csv
Safety & Trust		Toxicity analysis, anomaly detection, safety scoring	conversations_clean.csv, daily_kpis.csv, wildchat_combined_master.csv

1.2 Complete Visualization Inventory

Below is the full list of all 23 charts, KPI panels, and tables across all three dashboards:

#	Visualization Name	Chart Type	Dashboard Tab
1	KPI Strip — 5 Cards	KPI Tiles	Executive Summary
2	Weekly Engagement & Drop-off Trend	Dual-Axis Line Chart	Executive Summary
3	Prompt Category Mix	Donut Chart	Executive Summary
4	Top Geographies	Custom Table + Mini-Bar	Executive Summary
5	Model Version Performance	Grouped Bar Chart	Executive Summary
6	Platform Health Radar	Radar / Spider Chart	Executive Summary
7	Mini KPI Row — 4 Cards	KPI Tiles	Operational Intelligence
8	Power User Concentration	Pareto / Combo Bar+Line	Operational Intelligence
9	Conversation Funnel	Funnel / Horizontal Bar	Operational Intelligence
10	Activity Heatmap	Heatmap (Day × Hour)	Operational Intelligence
11	Prompt Length vs Response Quality	Scatter Plot	Operational Intelligence
12	Quality Score Distribution by Language	Box-and-Whisker Plot	Operational Intelligence
13	Prompt Category Trends Over Time	Stacked Area Chart	Operational Intelligence
14	Session Duration Distribution	Histogram	Operational Intelligence
15	User Cluster Explorer	Scatter Plot (K-Means)	Operational Intelligence
16	Response Quality by Language	Horizontal Bar	Operational Intelligence
17	Turn-by-Turn Retention Curves	Multi-Line Chart	Operational Intelligence

#	Visualization Name	Chart Type	Dashboard Tab
18	Language Distribution	Donut / Horizontal Bar	Operational Intelligence
19	Toxicity Rate + Z-Score Anomaly	Dual-Axis Line + Markers	Safety & Trust
20	Top Unsafe Prompt Categories	Horizontal Bar (Sorted)	Safety & Trust
21	Toxicity Concentration Heatmap	Heatmap	Safety & Trust
22	Sentiment Distribution by Category	Diverging Bar Chart	Safety & Trust
23	Safety Score Rolling Average	Line Chart	Safety & Trust
24	Policy Violation Category Trends	Stacked Bar Chart	Safety & Trust
25	Toxicity Rate by User Cluster	Bar Chart	Safety & Trust
26	Safety Score by Hour of Day	Line Chart	Safety & Trust

2. Dataset Description & Field Mapping

The dashboard uses six CSV files as data sources. Each file contains specific fields that map to particular visualizations. This section describes each file, its fields, and how they are used.

2.1 conversations_clean.csv

Primary dataset — 49,550 rows. One row per conversation. This is the main data source for most visualizations.

Field Name	Data Type	Role	Used In
conversation_id	String (ID)	Dimension	All views — primary key
conv_turn_count	Integer	Measure	KPI cards, retention curves, funnel
avg_prompt_len	Decimal	Measure	Scatter plot (X-axis)
avg_sentiment	Decimal (-1 to +1)	Measure	Sentiment distribution chart
country	String	Dimension	Geo table, map view
language	String	Dimension	Language distribution, quality by language
model	String	Dimension	Model performance bar chart
hashed_ip	String (ID)	Dimension	User/session counting
toxicity_score	Decimal (0–1)	Measure	Toxicity KPIs, heatmap, anomaly chart
toxic_flag	Integer (0/1)	Dimension/Measure	Toxicity rate calculation
drop_off_flag	Integer (0/1)	Dimension/Measure	Drop-off KPI, funnel
response_quality_score	Decimal	Measure	Quality by language, scatter Y-axis
prompt_category	String	Dimension	Donut chart, category trends, retention

2.2 daily_kpis.csv

Time-series dataset — 23 rows (one per day). Used for trend and anomaly charts.

Field Name	Data Type	Role	Used In
timestamp	Date	Dimension (Time)	Trend lines X-axis, rolling average
conversation_count	Integer	Measure	Engagement trend line
avg_tokens	Decimal	Measure	Operational performance chart
toxicity_rate	Decimal (0–1)	Measure	Toxicity anomaly overlay, safety score

2.3 geo_summary.csv

Country-level aggregated dataset — 149 rows. One row per country.

Field Name	Data Type	Role	Used In
country	String	Dimension	Geography table, map
conversation_count	Integer	Measure	Session volume bar, ranking
toxicity_rate	Decimal (0–1)	Measure	Geo toxicity column, color coding

2.4 messages_sample.csv

Message-level dataset — 554 rows. One row per message. Used for language and quality analysis.

Field Name	Data Type	Role	Used In
conversation_id	String (ID)	Dimension	Join key
message_index	Integer	Measure	Turn ordering
role	String (user/assistant)	Dimension	Funnel, retention
content	String (Text)	Dimension	Content analysis
timestamp	DateTime	Dimension (Time)	Hourly heatmap, time analysis
country	String	Dimension	Geo breakdown
language	String	Dimension	Language distribution donut
toxic	Integer (0/1)	Measure	Toxicity calculations
token_count	Integer	Measure	Scatter plot X-axis (prompt length)
sentiment_score	Decimal (-1 to +1)	Measure	Sentiment chart, box plots

2.5 prompt_categories.csv

Category summary — 11 rows. Used for prompt category breakdown.

Field Name	Data Type	Role	Used In
prompt_category	String	Dimension	Category labels in donut, bar, trend charts
count	Integer	Measure	Category volume sizing

2.6 wildchat_combined_master.csv

Master combined dataset — 49,550 rows. Combines conversations_clean and daily_kpis fields. Primary source for Safety & Trust dashboard.

Fields are a superset of conversations_clean.csv plus daily_kpis fields: date, conversation_count, avg_tokens, toxicity_rate.

3. Design System — Colors, Fonts & Layout

This section documents all colors, fonts, and layout rules used in the HTML dashboard. You must apply these EXACTLY in Tableau to match the original design.

3.1 Complete Color Palette (HEX Codes)

The dashboard uses the standard Tableau 10 categorical palette with specific background and text colors:

Color Name	HEX Code	Usage in Dashboard
Blue	#4E79A7	Primary charts, KPI accents, headers
Orange	#F28E2B	Drop-off KPI, warnings, secondary axis
Red	#E15759	Toxicity, anomalies, critical alerts
Teal	#76B7B2	Complementary data series
Green	#59A14F	Positive KPIs, On-Track status
Yellow	#EDC948	Supplementary data, medium risk
Purple	#B07AA1	Repeat user, creative category
Pink	#FF9DA7	Supplementary series
Brown	#9C755F	Tertiary data
Gray	#BAB0AC	Neutral / Other category
Header BG	#2D3E50	Top navigation bar background
Page BG	#F5F5F5	Dashboard page background
Panel BG	#FFFFFF	Chart & KPI card backgrounds
Text Primary	#1A1A2E	Main headings and labels
Text Muted	#5A6072	Subtitles, descriptions

3.2 Font Specifications

Font Role	Font Family	Size	Weight	Color
Dashboard Title (Topbar)	Source Serif 4 / Georgia	18px	Semibold 600	#FFFFFF on #2D3E50 bg
KPI Large Value	Source Serif 4 / Georgia	48px (hero), 34px	Semibold 600	#1A1A2E
Mini KPI Value	Source Serif 4 / Georgia	28px	Semibold 600	Accent color
Tab Navigation Labels	Source Sans 3 / Segoe UI	12px	Bold 700	#9A9FB0 (inactive), #4E79A7 (active)
Panel/Chart Titles	Source Sans 3 / Segoe UI	12px	Bold 700	#1A1A2E (uppercase)
KPI Card Labels	Source Sans 3 / Segoe UI	10px	Bold 700	#9A9FB0 (uppercase)

Font Role	Font Family	Size	Weight	Color
Body Text / Descriptions	Source Sans 3 / Segoe UI	11px	Regular 400	#5A6072
Axis Labels	Source Sans 3 / Segoe UI	11px	Regular 400	#9A9FB0
Table Data	Source Sans 3 / Segoe UI	12px	Regular 400	#1A1A2E
Status Tags / Badges	Source Sans 3 / Segoe UI	9-10px	Bold 700	Varies by status

📌 **NOTE:** In Tableau, use "Segoe UI" or "Arial" as substitutes for "Source Sans 3". Use "Georgia" as a substitute for "Source Serif 4" for KPI values.

3.3 Layout Grid Specifications

Each dashboard page uses a structured grid layout with the following containers:

Container Type	CSS Class	Description
Full Width KPI Row	.kpi-strip	5 columns: 2.2fr + 1fr + 1fr + 1fr + 1fr (hero card + 4 equal)
Mini KPI Row	.mini-kpi-row	4 equal columns, used in Operational dashboard
Two-Column Grid (7/5)	.g-7-5	7fr left panel + 5fr right panel
Equal Two-Column Grid	.g-6-6	1fr + 1fr equal split
Two-Column Grid (5/7)	.g-5-7	5fr left + 7fr right
Three-Column Grid	.g-3	3 equal columns
Four-Column Grid	.g-4	4 equal columns
Page Background	body	#F5F5F5 — applies to entire dashboard background
Card/Panel Background	.panel / .kpi-card	#FFFFFF with 1px solid #E0E0E0 border, 6px border-radius
Top Navigation Bar	.topbar	#2D3E50, 54px tall, sticky

3.4 KPI Card Design Rules

- Top accent bar: 3px colored stripe at the top of each KPI card
- Label text: 10px, UPPERCASE, letter-spacing 0.1em, color #9A9FB0
- Value text: Georgia/Serif font, 34px regular cards / 48px hero card
- Status tags below value: colored badge with icon (↑↓) and text
- Description note: 11px body text, color #5A6072
- Card padding: 18px 20px (regular) / 22px 24px (hero)

4. Setting Up Tableau — Data Import

Follow these steps to import all six data sources into Tableau Desktop before building any charts.

4.1 Open Tableau and Connect to Data

Step 1: Launch Tableau Desktop (version 2023.1 or newer recommended).
Step 2: On the Start screen, under "Connect", click "Text File".
Step 3: Navigate to your folder containing the CSV files.
Step 4: Select "conversations_clean.csv" and click Open.
Step 5: Tableau will open the Data Source page. Review the preview at the bottom.

4.2 Verify Field Types for conversations_clean.csv

After connecting, click on each field name in the data grid to confirm or correct the data type:

Field Name	Expected Type	How to Check / Fix
conversation_id	String (Abc)	Should auto-detect as text. If shown as number, click the # icon → Change to String.
conv_turn_count	Number (Integer)	Click # → verify "Number (whole)"
avg_prompt_len	Number (Decimal)	Click # → verify "Number (decimal)"
avg_sentiment	Number (Decimal)	Values range -1 to +1. Must be decimal.
country	String (Abc)	Right-click → Geographic Role → Country/Region
language	String (Abc)	Text dimension
model	String (Abc)	Text dimension — GPT model versions
hashed_ip	String (Abc)	Keep as string — do not convert to number
toxicity_score	Number (Decimal)	Values 0.0 to 1.0
toxic_flag	Number (Integer)	0 or 1 binary flag
drop_off_flag	Number (Integer)	0 or 1 binary flag
response_quality_score	Number (Decimal)	Quality score — decimal
prompt_category	String (Abc)	Category labels: Coding, Factual, Creative, etc.

4.3 Add Remaining Data Sources

Add each remaining CSV as a separate connection:

Step 1: Click "Add" button in the top-left of the Data Source page (next to connections list).
Step 2: Select "Text File" again and open "daily_kpis.csv".

Step 3: Repeat for: geo_summary.csv, messages_sample.csv, prompt_categories.csv, wildchat_combined_master.csv.

4.4 Set Geographic Role for Country Fields

Step 1: In the Data pane, right-click on the "country" field.

Step 2: Select "Geographic Role" → "Country/Region".

Step 3: Repeat this for the "country" field in geo_summary.csv and messages_sample.csv.

TIP: Setting the geographic role enables Tableau map views and latitude/longitude auto-generation.

4.5 Rename Data Sources for Clarity

In the Data pane (left side of a worksheet), right-click each source and rename:

- conversations_clean.csv → Conversations
- daily_kpis.csv → Daily KPIs
- geo_summary.csv → Geo Summary
- messages_sample.csv → Messages
- prompt_categories.csv → Prompt Categories
- wildchat_combined_master.csv → Master Dataset

5. Calculated Fields Reference

Calculated fields transform raw data into the KPI values and derived metrics shown in the dashboard. Create each field before building charts that use it.

How to Create a Calculated Field

Step 1: Open any Worksheet (click a tab at the bottom).

Step 2: From the top menu, click Analysis → Create Calculated Field...

Step 3: Type the field name in the "Name" box (exactly as shown below).

Step 4: Enter the formula in the formula box.

Step 5: Click OK. The new field appears in the Data pane under "Measures" (if numeric) or "Dimensions" (if string).

📌 **NOTE:** Create these calculated fields while connected to "Conversations" (conversations_clean.csv) as the primary data source, unless otherwise noted.

5.1 Engagement & Drop-off Calculated Fields

- Field: **User Engagement Rate** *(Source: Conversations (conversations_clean.csv))*
  Formula:
  `AVG(IF [conv_turn_count] > 2 THEN 1 ELSE 0 END)`
  Purpose: Returns 1 for conversations with more than 2 turns. AVG gives the engagement rate as a percentage.
  Used in: Executive Summary KPI Card (38%)
- Field: **Drop-off Rate** *(Source: Conversations)*
  Formula:
  `AVG(IF [drop_off_flag] = 1 THEN 1 ELSE 0 END)`
  Purpose: Calculates the proportion of conversations that ended within 2 turns (drop_off_flag = 1).
  Used in: KPI Card (62%), Funnel chart
- Field: **Avg Turn Count** *(Source: Conversations)*
  Formula:
  `AVG([conv_turn_count])`
  Purpose: Simple average of turn count across all conversations.
  Used in: KPI Card (3.4)
- Field: **Toxicity Rate** *(Source: Conversations)*
  Formula:
  `AVG([toxicity_score])`
  Purpose: Average toxicity score across all conversations. Multiply by 100 for percentage display.
  Used in: KPI Card (4.2%), Anomaly overlay
- Field: **Repeat User Rate** *(Source: Conversations)*
  Formula:
  `COUNT([conversation_id]) / COUNTD([hashed_ip])`
  Purpose: Number of conversations per unique IP. High ratio = repeat users. Normalize to 0-100% range.
  Used in: KPI Card (31%)
- Field: **Engagement Rate %** *(Source: Conversations)*

Formula:
```
[User Engagement Rate] * 100
```
Purpose: Converts the 0-1 rate to a display percentage.
Used in: KPI formatting

- Field: **Is Long Session** *(Source: Conversations)*

  Formula:
  ```
  IF [conv_turn_count] >= 8 THEN "Long Session" ELSE "Short Session" END
  ```
  Purpose: Classifies sessions for the long session uplift metric.
  Used in: Scatter plot color, operational KPI

- Field: **Toxicity Flag** *(Source: Conversations)*

  Formula:
  ```
  IF [toxicity_score] > 0.1 THEN 1 ELSE 0 END
  ```
  Purpose: Binary flag: 1 if conversation toxicity score exceeds 0.1 threshold.
  Used in: Toxicity rate heatmap, safety charts

- Field: **Quality Band** *(Source: Conversations)*

  Formula:
  ```
  IF [response_quality_score] >= 9 THEN "High" ELSEIF [response_quality_score] >= 6 THEN
  "Medium" ELSE "Low" END
  ```
  Purpose: Categorizes response quality into High/Medium/Low bands for color coding.
  Used in: Quality by language chart

- Field: **Z-Score Toxicity** *(Source: Daily KPIs (daily_kpis.csv))*

  Formula:
  ```
  ([toxicity_rate] - WINDOW_AVG([toxicity_rate])) / WINDOW_STDEV([toxicity_rate])
  ```
  Purpose: Standardized Z-score for anomaly detection. Values above 2.0 indicate anomalies.
  Used in: Anomaly overlay chart (secondary axis)

- Field: **Is Anomaly** *(Source: Daily KPIs)*

  Formula:
  ```
  IF [Z-Score Toxicity] > 2.0 THEN "Anomaly" ELSE "Normal" END
  ```
  Purpose: Flags daily records where toxicity Z-score exceeds 2 standard deviations.
  Used in: Anomaly markers on toxicity chart

- Field: **Rolling 7-Day Toxicity** *(Source: Daily KPIs)*

  Formula:
  ```
  WINDOW_AVG([toxicity_rate], -6, 0)
  ```
  Purpose: Calculates a 7-day rolling average of toxicity rate looking back 6 days.
  Used in: Safety score rolling average chart

- Field: **Category Share %** *(Source: Prompt Categories (prompt_categories.csv))*

  Formula:
  ```
  [count] / TOTAL(SUM([count])) * 100
  ```
  Purpose: Calculates the percentage share of each prompt category for the donut chart.
  Used in: Prompt category donut chart

- Field: **Cumulative Traffic Share** *(Source: Conversations)*

  Formula:
  ```
  RUNNING_SUM([User Traffic %])
  ```
  Purpose: Running cumulative sum for Pareto chart line.
  Used in: Power user Pareto chart (secondary axis)

6. DASHBOARD 1: Executive Summary

The Executive Summary dashboard is the first tab. It provides a high-level view of platform health including KPIs, engagement trends, geography, and model performance.

Page Layout Structure

Build the layout in this exact order from top to bottom:

1. Filter Bar (Language / Model / Date range toggles)
2. Section Header: "Platform Overview — KPIs · Trends · Geography · Model Performance"
3. KPI Strip — 5 cards in one horizontal row
4. Two-Column Grid (7/5): Engagement Trend Chart (left) + Platform Health Radar (right)
5. Two-Column Grid (6/6): Prompt Category Donut (left) + Model Performance Bar (right)
6. Full-Width: Top Geographies Table

6.1 KPI Strip — 5 Cards

Purpose: Show the 5 most important platform metrics at a glance.

6.1.1 Create Each KPI Worksheet

In Tableau, each KPI card is a separate worksheet. Create 5 worksheets:

KPI Card 1: User Engagement Rate (Hero Card)

Step 1: Create a new worksheet. Name it "KPI - Engagement Rate".

Step 2: Drag the calculated field "Engagement Rate %" to the Text shelf on the Marks card.

Step 3: Click on the Text mark on the Marks card → click "…" to open text editor.

Step 4: Format the number: right-click the field → Format → Number → Custom → "##.#%"

Step 5: In the Marks card dropdown (currently says "Automatic"), select "Text" chart type.

Step 6: Format the text: click the Text mark → set font to Georgia, size 48, bold, color #1A1A2E.

Step 7: Hide the axes: right-click the column header → uncheck "Show Header".

Step 8: Add a row: Drag a blank Paragraph or use a text annotation for the label "USER ENGAGEMENT RATE".

📌 **NOTE:** For the KPI label below the value, create a Caption text box in the worksheet with the description: "62% of conversations end within 2 turns."

KPI Card 2: Avg Turn Count

Step 1: Create a new worksheet: "KPI - Avg Turn Count".

Step 2: Drag "Avg Turn Count" (calculated field) to Text on Marks card.

Step 3: Format number: 1 decimal place → displays "3.4".

Step 4: Font: Georgia, size 34, bold, color #1A1A2E.

Step 5: Top accent color for this card will be GREEN (#59A14F) — set as background color in dashboard assembly.

KPI Card 3: Toxicity Rate

Step 1: Create worksheet: "KPI - Toxicity Rate".

Step 2: Drag "Toxicity Rate" (calculated field) to Text.

Step 3: Format: 1 decimal place with % symbol → "4.2%".

Step 4: Font: Georgia, size 34, color #1A1A2E. Top accent = RED (#E15759).

KPI Card 4: Drop-off Rate

Step 1: Create worksheet: "KPI - Drop-off Rate".

Step 2: Drag "Drop-off Rate" × 100 to Text.

Step 3: Format: Integer with % → "62%". Top accent = ORANGE (#F28E2B).

KPI Card 5: Repeat User Rate

Step 1: Create worksheet: "KPI - Repeat User Rate".

Step 2: Create a formula or use COUNTD to derive the repeat rate.

Step 3: Format: Integer with % → "31%". Top accent = PURPLE (#B07AA1).

6.1.2 KPI Card Accent Bars

Each KPI card has a 3px colored bar at the top. Simulate this in Tableau:

Step 1: In the dashboard, add a Horizontal Container above each KPI worksheet.

Step 2: Inside it, add a Blank object with a fixed height of 5px.

Step 3: Right-click the blank object → Edit Background → set background color to the card's accent color.

TIP: Alternatively, add a colored border to the top of the KPI worksheet: Format → Borders → Row Divider → set to the accent HEX color with thickness 3.

6.2 Weekly Engagement & Drop-off Trend — Dual-Axis Line Chart

Data source: daily_kpis.csv | Chart type: Dual-Axis Line | Size: 7 parts of the left column

Step-by-Step Build Instructions

Step 1: Create a new worksheet. Name it "Engagement Trend".

Step 2: Ensure "Daily KPIs" is the selected data source (click it in the Data pane).

Step 3: Drag [timestamp] to Columns. Right-click it → select "Week" (not exact date). This creates weekly aggregation.

Step 4: Drag [conversation_count] to Rows. This creates the first line.

Step 5: Drag [toxicity_rate] to Rows (same row shelf, second pill). This creates a second axis.

Step 6: Right-click the second axis (right side) → "Synchronize Axis". Then right-click again → "Dual Axis".

Step 7: On the Marks card, click the first measure mark (conversation_count) → change type to "Line".

Step 8: On the second mark (toxicity_rate) → also set to "Line".

Step 9: Click the Color shelf on the first mark → change to #4E79A7 (blue).

Step 10: Click the Color shelf on the second mark → change to #F28E2B (orange).

Step 11: Set line thickness: right-click each line → Format → Line → Width → 2px.

Step 12: Add reference lines: Right-click the left axis → Add Reference Line → set to average. Style: dashed, color matching the line.

Formatting the Chart

Step 1: Hide grid lines: Format menu → Lines → Grid Lines → set to None.

Step 2: Axis titles: Double-click left axis → type "Conversations / Week". Right axis → "Toxicity Rate (%)".

Step 3: Chart title: Double-click the title area → type "Weekly Engagement & Drop-off Trend".

Step 4: Set title font: Arial, 12px, Bold, Uppercase, color #1A1A2E.

Step 5: Background: Format → Shading → Worksheet → White (#FFFFFF).

Step 6: Tooltip: click Tooltip shelf → customize to show "Week: [WEEK(timestamp)]" and both metric values.

6.3 Prompt Category Mix — Donut Chart

Data source: prompt_categories.csv | Chart type: Donut (Pie with hole) | Colors: Tableau 10 palette

Step 1: Create a new worksheet: "Category Donut".

Step 2: Select "Prompt Categories" data source.

Step 3: Drag [count] to Columns.

Step 4: Drag [prompt_category] to Color on the Marks card.

Step 5: In the Marks card dropdown, select "Pie".

Step 6: Move [count] from Columns to the Angle shelf on the Marks card (this creates the pie slices).

Step 7: Drag [prompt_category] to the Label shelf. Format label: show category name + percentage.

Creating the Donut Hole

Step 1: Hold Ctrl and drag the Angle measure to create a duplicate. Now you have two SUM([count]) pills on Rows.

Step 2: Right-click the second axis → Dual Axis.

Step 3: On the second Marks card, remove everything from shelves (clear Color, Label, Angle).

Step 4: Change the second Marks type to "Pie".

Step 5: Set the second pie's size to about 50% by clicking Size on the Marks card and reducing it.

Step 6: Set the second pie Color to White (#FFFFFF). This creates the donut hole.

Apply Category Colors

Click Color on the Marks card → Edit Colors. Assign these exact HEX values:

Category	HEX Color
Coding & Technical	#4E79A7
Factual Q&A	#6BAED6
Creative Writing	#B07AA1
Emotional Support	#59A14F
Roleplay / Persona	#F28E2B
Harmful / Violating	#E15759
Other	#BAB0AC

6.4 Top Geographies Table

Data source: geo_summary.csv | Chart type: Text Table with embedded mini-bars | Shows top 10 countries

Step 1: Create a new worksheet: "Geo Table".

Step 2: Select "Geo Summary" data source.

Step 3: Drag [country] to Rows.

Step 4: Drag [conversation_count] to Columns → right-click → Sort Descending. This ranks countries.

Step 5: Drag [conversation_count] to the Text shelf on Marks. Drag [toxicity_rate] also to Text.

Step 6: Change Marks type to "Text".

Step 7: Right-click the Count row → Filter → Top 10 by [conversation_count].

Adding Mini Progress Bars (Bar-in-Cell)

Step 1: To add the mini bar next to conversation_count, drag [conversation_count] to Columns a second time.

Step 2: Right-click the second [conversation_count] → Dual Axis.

Step 3: Change the second marks card type to "Bar".

Step 4: Set bar color to #4E79A7 (blue). Reduce bar size to Small.

Step 5: Hide the second axis: right-click it → uncheck "Show Header".

Step 6: Format table borders: Format → Borders → reduce to minimal (show only row dividers at 1px #E0E0E0).

Table Column Headers

Add the following column aliases (right-click each field → Rename):

- [country] → "Country"
- [conversation_count] → "Sessions"
- [toxicity_rate] → "Tox %"

6.5 Model Version Performance — Grouped Bar Chart

Data source: conversations_clean.csv | Chart type: Horizontal Bar Chart | Shows GPT model comparison

Step 1: Create a new worksheet: "Model Performance".

Step 2: Drag [model] to Rows.

Step 3: Drag [response_quality_score] to Columns. Change aggregation: right-click → Measure → Average.

Step 4: Drag [toxicity_score] to Columns. Aggregation: Average.

Step 5: In the Marks card, set type to "Bar".

Step 6: Drag [model] to Color shelf to color bars by model.

Step 7: Sort by quality score descending: right-click the left axis → Sort → Field → Descending → [response_quality_score].

Step 8: Add data labels: drag [response_quality_score] to Label shelf → format to 1 decimal place.

Formatting Model Performance Chart

Step 1: Set bar colors: gpt-3.5-turbo → #4E79A7 (blue), gpt-4 models → #59A14F (green), others → #BAB0AC.

Step 2: Remove grid lines: Format → Lines → Grid Lines → None.

Step 3: Title: "Model Version Performance" — Arial, 12px, Bold, Uppercase.

6.6 Platform Health Radar Chart

Data source: Manual / Dashboard-level | Chart type: Radar (requires reshaping)

The radar chart shows 6 platform dimensions: Engagement, Drop-off (Inverted), Quality, Retention, Safety, and Multilingual. It compares Actual vs Target values.

Preparing the Radar Data

Step 1: Create a new CSV file named "radar_data.csv" with this content:

```
Dimension,Actual,Target
Engagement,68,80
Drop-off (Inv),38,20
Quality,72,85
Retention,58,75
Safety,85,90
Multilingual,42,60
```

Step 2: Connect this new CSV file to Tableau as a 7th data source named "Radar Data".

Step 3: Reshape using Tableau Pivot: select both [Actual] and [Target] columns → right-click → Pivot. Rename the pivot field names to "Metric" and "Score".

Building the Radar/Spider Chart in Tableau

Tableau does not have a native radar chart. Use the polygon approach:

Step 1: Create calculated fields for X and Y coordinates for each dimension using trigonometry.

Step 2: The formula for X: COS(INDEX()/6 * 2 * PI()) * [Score]

Step 3: The formula for Y: SIN(INDEX()/6 * 2 * PI()) * [Score]

Step 4: Drag X to Columns, Y to Rows.

Step 5: Change Marks type to "Polygon" and drag [Metric] to Path shelf.

Step 6: Drag [Actual vs Target] (after pivot) to Color shelf. Set Actual = #4E79A7, Target = dashed #BAB0AC.

TIP: Alternatively, use a Spider Chart extension from Tableau Exchange if trigonometric approach is complex.

7. DASHBOARD 2: Operational Intelligence

The Operational Intelligence tab provides deep analysis of user behavior, conversation patterns, and segment-level insights. It contains the most charts (12 visualizations) and has an activity heatmap toggle.

Page Layout Structure

7. Anomaly Alert Strip (red banner at top)
8. Mini KPI Row — 4 equal cards
9. Two-Column Grid (7/5): Category Trends Area Chart (left) + Conversation Funnel (right)
10. Full-Width: Activity Heatmap (toggle between Volume and Toxicity views)
11. Two-Column Grid (6/6): Scatter Plot Prompt Length vs Quality (left) + Box Plot Language Quality (right)
12. Two-Column Grid (5/7): Session Duration Histogram (left) + User Cluster Scatter (right)
13. Two-Column Grid (6/6): Response Quality by Language (left) + Retention Curves (right)
14. Two-Column Grid (5/7): Language Distribution Donut/Bar (left) + Power User Pareto (right)

7.1 Mini KPI Row — 4 Cards

Four narrow KPI cards, each with a colored left-side accent bar.

Card #	Label	Value	Accent Color	Formula
1	Drop-off · Turn 2	62%	#F28E2B (Orange)	AVG([drop_off_flag]) * 100
2	Power User Share	8%	#4E79A7 (Blue)	COUNTD top 8% of [hashed_ip] by turn count
3	Long Session Uplift	+35%	#59A14F (Green)	Hardcoded label, derived metric
4	Language Gap	2.4×	#E15759 (Red)	Non-English unanswered rate / English rate

Build each card following the same instructions as Section 6.1, but use the left-side accent (3px colored left border) instead of top accent.

7.2 Power User Concentration — Pareto Chart

Data source: conversations_clean.csv | Chart type: Bar + Cumulative Line (Pareto)

Step 1: Create worksheet: "Power User Pareto".

Step 2: Create a calculated field "User Decile": NTILE(10) of SUM([conv_turn_count]) partitioned by [hashed_ip]. This splits users into 10 equal groups.

Step 3: Drag [User Decile] to Columns. Right-click → Discrete.

Step 4: Drag COUNT([conversation_id]) to Rows (or [conv_turn_count] as sum).

Step 5: In Marks card, change type to "Bar". Color bars #4E79A7.

Step 6: Drag a second measure (cumulative share) to Rows. Set as dual axis.

Step 7: Change second marks card to "Line". Color: #F28E2B (orange).

Step 8: Right-click second axis → Synchronize → make it 0–100%.

Step 9: Add reference line at 80% on cumulative line: right-click → Add Reference Line → Constant → 80.

Step 10: Label the bars with their traffic % using Label shelf.

7.3 Conversation Funnel Chart

Data source: conversations_clean.csv | Chart type: Horizontal Bar (Funnel simulation)

Step 1: Create worksheet: "Conversation Funnel".

Step 2: Create a calculated field "Turn Cohort" based on turn count brackets:

```
IF [conv_turn_count] >= 1 THEN "1. Session Started" END
IF [conv_turn_count] >= 2 THEN "2. Turn 2 Reached" END
-- etc. through "5. Power Session (8+ Turns)"
```

Step 3: Count conversations in each cohort using LOD expressions:

```
{ FIXED : COUNTD(IF [conv_turn_count] >= 1 THEN [conversation_id] END) }
```

Step 4: Drag the stage labels to Rows and counts to Columns.

Step 5: Make bars symmetric (funnel shape): create a negative mirror bar.

Step 6: Apply gradient blue colors: darkest = Stage 1, lightest = final stage.

TIP: Use a pre-aggregated helper table if LOD expressions are complex. Create a separate sheet with turn count bins.

7.4 Activity Heatmap — Day of Week × Hour of Day

Data source: messages_sample.csv | Chart type: Heatmap | Toggle: Volume / Toxicity

Step 1: Create worksheet: "Activity Heatmap".

Step 2: Drag [timestamp] to Columns. Right-click → More → Hour. Rename column alias "Hour of Day".

Step 3: Drag [timestamp] to Rows. Right-click → More → Day of Week. This gives Mon–Sun rows.

Step 4: On Marks card, change type to "Square".

Step 5: Drag COUNT([conversation_id]) to Color shelf.

Step 6: Click Color → Edit Colors → choose a sequential blue palette (Tableau Blue-Teal).

Step 7: Set minimum color = #EEF4FB, maximum = #2D3E50.

Step 8: Drag SIZE to make marks fill the cells: right-click → Entire View.

Step 9: Format borders: Format → Cell Size → Square to make cells equal.

Adding the Toxicity View Toggle

In Tableau, create two separate heatmap worksheets and use a Parameter to switch between them:

Step 1: Create a Parameter: Data → Create Parameter → Name "Heatmap Mode" → Type String → List: "Volume", "Toxicity".

Step 2: Create a calculated field: IF [Heatmap Mode] = "Volume" THEN COUNT([conversation_id]) ELSE AVG([toxicity_score]) END

Step 3: Use this field on the Color shelf instead of the raw count.

Step 4: Add a Parameter Control to the dashboard: right-click the parameter → "Show Parameter".

7.5 Prompt Length vs Response Quality — Scatter Plot

Data source: conversations_clean.csv | Chart type: Scatter Plot | X: avg_prompt_len | Y: response_quality_score

Step 1: Create worksheet: "Prompt Length Scatter".

Step 2: Drag [avg_prompt_len] to Columns. Aggregation: AVG or keep as dimension by right-clicking → Dimension.

Step 3: Drag [response_quality_score] to Rows. Aggregation: AVG.

Step 4: Drag [conversation_id] to Detail shelf to show individual points.

Step 5: Drag [prompt_category] to Color shelf. Apply the category color palette from Section 6.3.

Step 6: In Marks card, type should be "Automatic" or "Circle".

Step 7: Set opacity: click Color shelf → set Opacity to 70%.

Step 8: Add trend line: Analysis menu → Trend Lines → Show Trend Lines.

Step 9: Format trend line: right-click → Edit Trend Lines → choose Linear, color #9A9FB0, dashed.

7.6 Quality Score Distribution by Language — Box Plot

Data source: conversations_clean.csv / messages_sample.csv | Chart type: Box-and-Whisker Plot

Step 1: Create worksheet: "Quality Box Plot".

Step 2: Drag [language] to Columns. Filter to Top 8 languages by count.

Step 3: Drag [response_quality_score] to Rows.

Step 4: In Marks card, change type to "Circle" (individual data points).

Step 5: From the Analysis menu → Box Plot → add Box Plots.

Step 6: This overlays the box and whiskers on the circles.

Step 7: Color the circles by language: drag [language] to Color.

Step 8: Format box: right-click box plot → Format → set fill to light color, border to accent color.

7.7 Prompt Category Trends — Stacked Area Chart

Data source: conversations_clean.csv | Chart type: Stacked Area | X: date | Y: count by category

Step 1: Create worksheet: "Category Trends".
Step 2: Drag [date] or [timestamp] to Columns. Set to Week level.
Step 3: Drag COUNT([conversation_id]) to Rows.
Step 4: Drag [prompt_category] to Color shelf.
Step 5: In Marks card, change type to "Area".
Step 6: Tableau automatically stacks the areas by the color dimension.
Step 7: Apply category colors (same as Section 6.3 donut chart).
Step 8: Add a filter for the last 26 weeks: right-click [date] → Filter → Relative date → Last 26 Weeks.

7.8 Session Duration Distribution — Histogram

Data source: conversations_clean.csv | Chart type: Histogram (bar chart with bins)

Step 1: Create worksheet: "Duration Histogram".
Step 2: Right-click [conv_turn_count] in the Data pane → Create → Bins.
Step 3: In the Bins dialog, set bin size = 2 (or use manual bins for 0–1, 1–3, 3–5, 5–10, etc.).
Step 4: Drag the new bin field to Columns.
Step 5: Drag COUNT([conversation_id]) to Rows.
Step 6: In Marks card, type = "Bar".
Step 7: Color: set bars to a blue gradient using a sequential custom color palette.
Step 8: Sort bars by bin order (they should be in ascending duration order automatically).
Step 9: Add labels: drag COUNT([conversation_id]) to Label shelf.

7.9 User Cluster Explorer — Scatter Plot

Data source: conversations_clean.csv | Chart type: Scatter Plot | Simulates K-Means clusters

Step 1: Create worksheet: "User Cluster".

Step 2: Create two calculated fields to represent cluster axes (proxy PCA dimensions):

```
Cluster Dim 1: [conv_turn_count] / WINDOW_MAX([conv_turn_count])
Cluster Dim 2: [response_quality_score] / WINDOW_MAX([response_quality_score])
```

Step 3: Drag Cluster Dim 1 to Columns, Cluster Dim 2 to Rows.

Step 4: Create a "User Cluster" calculated field using turn count and toxicity thresholds:

```
IF [conv_turn_count] > 10 AND [toxicity_score] < 0.05 THEN "Power Users"
ELSEIF [conv_turn_count] > 5 THEN "Task-Oriented"
ELSEIF [toxicity_score] > 0.15 THEN "Adversarial"
ELSEIF [drop_off_flag] = 1 THEN "At-Risk"
ELSE "Casual Explorers" END
```

Step 5: Drag [User Cluster] to Color shelf.

Step 6: Apply cluster colors: Power Users = #4E79A7, Task = #59A14F, Casual = #F28E2B, At-Risk = #E07050, Adversarial = #E15759.

7.10 Response Quality by Language — Horizontal Bar

Data source: conversations_clean.csv | Chart type: Horizontal Bar Chart

Step 1: Create worksheet: "Quality by Language".

Step 2: Drag [language] to Rows.

Step 3: Drag AVG([response_quality_score]) to Columns.

Step 4: Sort by quality descending.

Step 5: Color bars using [Quality Band] calculated field: High = #59A14F, Medium = #F28E2B, Low = #E15759.

Step 6: Add data labels on bars.

Step 7: Filter to top 10 languages by conversation count.

7.11 Turn-by-Turn Retention Curves — Multi-Line Chart

Data source: conversations_clean.csv | Chart type: Multi-Line Chart

Step 1: Create worksheet: "Retention Curves".

Step 2: Create a helper table showing retention by turn number for each category.

Step 3: Create a calculated field "Turn Number": this requires a reshaped dataset. Each row should represent a turn event.

Step 4: If reshaping is complex, create an aggregate file: for each (category, turn_number) pair, calculate the % of conversations still active.

Step 5: Drag [Turn Number] (1–8+) to Columns.

Step 6: Drag [Retention %] to Rows.

Step 7: Drag [prompt_category] to Color.

Step 8: Marks type = "Line".

Step 9: Apply category colors from Section 6.3.

Step 10: Add reference line at Turn 2 (X=2) showing the 62% average drop-off.

7.12 Language Distribution — Donut/Bar Toggle

Data source: messages_sample.csv | Chart type: Donut or Horizontal Bar (togglable)

Step 1: Create worksheet: "Language Distribution".

Step 2: Drag [language] to Color and Rows.

Step 3: Create the donut using the same technique as Section 6.3.

Step 4: Create a second worksheet "Language Bar" with [language] on Rows and COUNT on Columns.

Step 5: Use a Parameter control "Language View" with values "Donut" and "Bar".

Step 6: In dashboard, use sheet selector or show/hide based on parameter value.

Step 7: Top 10 languages by count, sorted descending.

8. DASHBOARD 3: Safety & Trust

The Safety & Trust dashboard focuses on toxicity monitoring, anomaly detection, and policy compliance. It features an anomaly alert banner and 8 safety-related visualizations.

Page Layout Structure

15. Anomaly Alert Banner (red strip at top — "TOXICITY SPIKE DETECTED")
16. Three KPI Cards: Toxicity Rate (4.2%), Rate vs Baseline (2.8×), Flagged Sessions
17. Two-Column Grid (7/5): Toxicity Anomaly Chart (left) + Top Unsafe Categories (right)
18. Two-Column Grid (6/6): Toxicity Heatmap (left) + Sentiment Distribution (right)
19. Two-Column Grid (7/5): Safety Rolling Average (left) + Active Anomaly Events list (right)
20. Two-Column Grid (6/6): Policy Violation Trends (left) + Toxicity by Cluster (right)
21. Full-Width: Safety by Hour of Day

8.1 Toxicity KPI Banner

Three KPI cards in a horizontal row showing: current toxicity rate, rate vs baseline, and flagged sessions count.

Step 1: Create three KPI worksheets following the same process as Section 6.1.
Step 2: KPI 1 — Toxicity Rate: AVG([toxicity_score]) × 100 → "4.2%" with red accent.
Step 3: KPI 2 — Rate vs Baseline: AVG([toxicity_score]) / 0.015 → "2.8×" with orange accent.
Step 4: KPI 3 — Flagged Sessions: COUNT(IF [toxic_flag] = 1 THEN [conversation_id] END) with red accent.

8.2 Toxicity Rate + Z-Score Anomaly Overlay

Data source: daily_kpis.csv | Chart type: Line (toxicity) + Dual Axis Line with markers (Z-score)

Step 1: Create worksheet: "Toxicity Anomaly".
Step 2: Drag [timestamp] to Columns. Format as Day level.
Step 3: Drag [toxicity_rate] to Rows. Change aggregation to AVG.
Step 4: Drag [Z-Score Toxicity] calculated field to Rows. Set as dual axis.
Step 5: Left axis (toxicity_rate): line chart, color #4E79A7 (blue), line width 2.
Step 6: Right axis (Z-Score): change marks type to "Line + Shapes" or "Area".
Step 7: Color the Z-score line: use a conditional color — above 2.0 = #E15759, below = #BAB0AC.
Step 8: Add anomaly markers: drag [Is Anomaly] calculated field to Shape shelf.
Step 9: For anomaly points, use a circle/dot shape, color #E15759, size Large.
Step 10: Add a reference band: right-click left axis → Add Reference Line → Band → from 0 to 3 (normal range).
Step 11: Format the reference band: fill color = light green #EAF4EA, border = dashed.

8.3 Top Unsafe Prompt Categories — Horizontal Bar

Data source: conversations_clean.csv or prompt_categories.csv | Chart type: Sorted Horizontal Bar

Step 1: Create worksheet: "Unsafe Categories".

Step 2: Filter [prompt_category] to only "Harmful / Violating" or create a sub-categorization.

Step 3: Create a calculated field "Unsafe Category":

```
IF CONTAINS([prompt_category], "Political") THEN "Political / Extremism"
ELSEIF CONTAINS([prompt_category], "Explicit") THEN "Explicit / Adult"
-- etc.
ELSE [prompt_category] END
```

Step 4: Drag [Unsafe Category] to Rows.

Step 5: Drag COUNT([conversation_id]) to Columns.

Step 6: Sort descending. Type = "Bar". Color = #E15759 (red) for all bars.

Step 7: Add data labels. Set font: Arial, 11px.

Step 8: Add a rate annotation showing "per 1K sessions" for each bar.

8.4 Toxicity Concentration Heatmap

Data source: conversations_clean.csv | Chart type: Heatmap by Country and Language

Step 1: Create worksheet: "Toxicity Heatmap".

Step 2: Drag [country] to Rows and [language] to Columns.

Step 3: Drag AVG([toxicity_score]) to Color shelf.

Step 4: Marks type = "Square".

Step 5: Edit colors: sequential orange-red scale. Min = white (#FFFFFF), Max = dark red (#B03030).

Step 6: Filter to Top 10 countries and Top 8 languages.

Step 7: Set cell size to "Entire View" for consistent square cells.

8.5 Sentiment Distribution by Category — Diverging Bar Chart

Data source: conversations_clean.csv | Chart type: Diverging Bar (positive/negative stacked)

Step 1: Create worksheet: "Sentiment Distribution".

Step 2: Create calculated fields for positive and negative sentiment:

```
Positive Sentiment: IF [avg_sentiment] > 0 THEN [avg_sentiment] ELSE 0 END
Negative Sentiment: IF [avg_sentiment] < 0 THEN [avg_sentiment] ELSE 0 END
```

Step 3: Drag [prompt_category] to Rows.

Step 4: Drag [Positive Sentiment] to Columns.

Step 5: Drag [Negative Sentiment] to Columns as a dual axis.

Step 6: Marks type = "Bar" for both.

Step 7: Positive bars: color #59A14F (green). Negative bars: color #E15759 (red).

Step 8: Add reference line at X=0 for the center baseline.

8.6 Safety Score Rolling Average

Data source: daily_kpis.csv | Chart type: Line Chart with 7-day rolling average

Step 1: Create worksheet: "Safety Rolling Average".
Step 2: Drag [timestamp] to Columns (Day level).
Step 3: Drag [toxicity_rate] to Rows.
Step 4: Add a second line: drag [Rolling 7-Day Toxicity] calculated field to Rows.
Step 5: Set dual axis. Synchronize.
Step 6: Raw toxicity line: thin, light gray #BAB0AC, opacity 50%.
Step 7: Rolling average line: thick 2px, orange #F28E2B.
Step 8: Mark anomaly points: use Shape on marks for the 3 spike dates (D09, D19, D26).
Step 9: Anomaly markers: red circle, size Large, color #E15759.

8.7 Policy Violation Category Trends — Stacked Bar

Data source: daily_kpis.csv + conversations_clean.csv | Chart type: Stacked Bar by week

Step 1: Create worksheet: "Violation Trends".
Step 2: Drag [timestamp] to Columns. Set to Week.
Step 3: Drag COUNT([conversation_id]) to Rows.
Step 4: Filter to only toxic conversations: right-click [toxic_flag] → Filter → select "1".
Step 5: Drag [Unsafe Category] to Color shelf.
Step 6: Marks type = "Bar". Tableau auto-stacks.
Step 7: Apply violation category colors:

- Political/Extremism: #E15759 (red)
- Explicit/Adult: #E07050 (dark orange)
- Self-Harm/Crisis: #F28E2B (orange)
- Hate Speech: #EDC948 (yellow)
- Other: #BAB0AC (gray)

8.8 Toxicity Rate by User Cluster

Data source: conversations_clean.csv | Chart type: Bar Chart sorted by toxicity rate

Step 1: Create worksheet: "Cluster Toxicity".
Step 2: Drag [User Cluster] calculated field to Rows.
Step 3: Drag AVG([toxicity_score]) × 100 to Columns.
Step 4: Sort descending by toxicity rate.
Step 5: Color bars by cluster color:

- Adversarial: #E15759 — highest (28.4%)
- At-Risk: #E07050 — second (7.1%)
- Casual: #F28E2B — third (3.2%)
- Power Users: #4E79A7 — fourth (2.8%)
- Task-Oriented: #59A14F — lowest (1.4%)

Step 6: Add data labels showing the exact toxicity rate.

8.9 Safety Score by Hour of Day

Data source: messages_sample.csv | Chart type: Line Chart X=Hour, Y=Toxicity Rate

Step 1: Create worksheet: "Safety by Hour".

Step 2: Drag [timestamp] to Columns → right-click → More → Hour → Discrete.

Step 3: Drag AVG([toxicity_score]) × 100 to Rows.

Step 4: Marks type = "Line + Area".

Step 5: Fill area below the line with semi-transparent red (#E15759 at 15% opacity).

Step 6: Color line: gradient from green (low risk hours 9–17) to red (high risk 22–02).

Step 7: Add reference band for "Safe Hours" (9:00–17:00): right-click axis → Reference Band → from 9 to 17 → fill with light green.

Step 8: Add annotations at hour 23:00 showing "Peak risk zone".

9. Dashboard Layout & Assembly

This section explains how to assemble all individual worksheets into the three full dashboards using Tableau's layout containers. This is the most important step for matching the HTML design exactly.

9.1 Understanding Tableau Layout Containers

Tableau uses two types of containers to organize dashboard elements:

Container Type	Use Case	How to Add
Horizontal Container	Place elements SIDE BY SIDE in a row (like the KPI strip, two-column grids)	Drag "Horizontal" from Objects panel (bottom-left in Dashboard view) to canvas
Vertical Container	Stack elements TOP TO BOTTOM (like the full dashboard column)	Drag "Vertical" from Objects panel to canvas
Blank Object	Add spacing/padding between elements	Drag "Blank" from Objects panel
Text Object	Add titles, descriptions, section headers	Drag "Text" from Objects panel → double-click to edit
Image Object	Add logos or decorative elements	Drag "Image" from Objects panel

9.2 Creating a New Dashboard

Step 1: At the bottom of the Tableau window, click the "New Dashboard" icon (looks like a grid with a +).

Step 2: The Dashboard view opens. You will see the canvas (center) and an Objects panel (left sidebar).

Step 3: Set dashboard size: in the left panel under "Size", select "Fixed size" → set to 1400 × 900 pixels.

Step 4: Background color: Format → Dashboard → set Shading to #F5F5F5.

9.3 Assembling Dashboard 1: Executive Summary

Follow this exact sequence of steps:

Top Navigation Bar (Topbar)

Step 1: Drag a "Text" object to the top of the canvas.

Step 2: Type the dashboard title: "WildChat Analytics Platform — Group G-6"

Step 3: Set background color to #2D3E50. Font: Georgia, 18px, white.

Step 4: Set fixed height to 54px.

Tab Navigation Row

Step 1: Add a "Horizontal Container" below the topbar.

Step 2: Inside it, add three "Navigation Button" objects (Dashboard → Objects → Navigation).

Step 3: Label them: "📊 Executive Summary", "⚙ Operational Intelligence", "🛡 Safety & Trust".

Step 4: Link each button to its corresponding Dashboard (you must create all 3 dashboards first to link them).

Step 5: Background: white (#FFFFFF). Active tab: add a bottom border line (workaround: use a colored Blank object).

Filter Bar

Step 1: Add a "Horizontal Container" below the tabs.

Step 2: Drag filter controls (from any worksheet) into this container.

Step 3: Filters to include: Language (multi-select), Model (multi-select), Date Range (relative).

Step 4: Background: white, border: 1px #E0E0E0, border-radius: not available in Tableau (use workaround with padding).

KPI Strip — 5 Cards in a Row

Step 1: Drag a "Horizontal Container" to the canvas below the filter bar.

Step 2: Inside the container, drag the 5 KPI worksheets in this order:

- "KPI - Engagement Rate" (Hero) — set width to approximately 25% of container
- "KPI - Avg Turn Count" — set to equal 25% width
- "KPI - Toxicity Rate" — equal width
- "KPI - Drop-off Rate" — equal width
- "KPI - Repeat User Rate" — equal width

Step 3: Right-click each KPI worksheet in the dashboard → "Fit" → "Entire View".

Step 4: Set container height to 140px (fixed height in the Layout panel).

Main Grid — Engagement Trend + Radar (7/5 split)

Step 1: Drag a "Horizontal Container" below the KPI strip.

Step 2: Drag the "Engagement Trend" worksheet into the left portion.

Step 3: Drag the "Platform Health Radar" worksheet into the right portion.

Step 4: In the Layout panel (right side), select the left worksheet → set Width to 58% (7 parts of 12).

Step 5: The right worksheet automatically fills the remaining 42% (5 parts of 12).

Step 6: Set container height to 300px.

Second Row — Category Donut + Model Performance (6/6 split)

Step 1: Drag another "Horizontal Container" below the trend row.

Step 2: Add "Category Donut" worksheet to the left (50%).

Step 3: Add "Model Performance" worksheet to the right (50%).

Step 4: Set container height to 280px.

Geography Table (Full Width)

Step 1: Drag the "Geo Table" worksheet to the canvas below the two-column row.

Step 2: Set it to full width. Set height to 250px.

9.4 Adding Panel Borders and Backgrounds

Each chart panel should look like a card with a white background and subtle border:

Step 1: Right-click each worksheet in the dashboard → "Format".

Step 2: In the Format panel → "Shading" → set Worksheet background to #FFFFFF.

Step 3: In Format → "Border" → set outer border to #E0E0E0, 1px.

Step 4: Add 8px padding inside each worksheet: Layout panel → Outer Padding → all sides 8.

📌 **NOTE:** Tableau does not support CSS border-radius. The card corners will be square, which is an acceptable difference from the HTML version.

9.5 Assembling Dashboard 2: Operational Intelligence

Follow the same process as Dashboard 1. The key layout differences are:

Row	Content	Container Type	Heights
Row 1	Anomaly Alert Banner (text box, red background)	Full-width Horizontal	50px
Row 2	4 Mini KPI Cards	Horizontal (4 equal)	110px
Row 3	Category Trends Area (7) + Funnel (5)	Horizontal (58/42%)	320px
Row 4	Activity Heatmap with Mode toggle	Full-width Horizontal	300px
Row 5	Scatter Plot (6) + Box Plot (6)	Horizontal (50/50%)	300px
Row 6	Duration Histogram (5) + User Cluster (7)	Horizontal (42/58%)	300px
Row 7	Quality by Language (6) + Retention Curves (6)	Horizontal (50/50%)	300px
Row 8	Language Distribution (5) + Power User Pareto (7)	Horizontal (42/58%)	300px

9.6 Assembling Dashboard 3: Safety & Trust

The Safety dashboard has a distinctive red-themed color scheme for anomaly elements:

Row	Content	Container Type	Heights
Row 1	Safety Alert Banner (red strip, "ANOMALY ACTIVE")	Full-width, red bg	50px
Row 2	3 Safety KPI Cards	Horizontal (3 equal)	130px
Row 3	Toxicity Anomaly Chart (7) + Top Unsafe Categories (5)	Horizontal (58/42%)	320px
Row 4	Toxicity Heatmap (6) + Sentiment Distribution (6)	Horizontal (50/50%)	300px
Row 5	Safety Rolling Average (7) + Anomaly Events List (5)	Horizontal (58/42%)	280px
Row 6	Violation Trends (6) + Toxicity by Cluster (6)	Horizontal (50/50%)	280px
Row 7	Safety Score by Hour of Day	Full-width Horizontal	260px

10. Filters, Interactions & Navigation

10.1 Adding Dashboard Filters

Filters that appear in the Filter Bar at the top of each dashboard:

Language Filter

Step 1: In any worksheet using conversations_clean.csv, drag [language] to the Filters shelf.

Step 2: In the filter dialog, select "All" to show all values, then click OK.

Step 3: Right-click the [language] filter → "Show Filter".

Step 4: The filter control appears on the right side of the worksheet. On the dashboard, drag it to the Filter Bar container.

Step 5: Right-click the filter control → "Apply to Worksheets" → "All Using This Data Source".

Model Filter

Step 1: Repeat the same process with the [model] field.

Step 2: Filter type: Dropdown or button list. In the filter control, click the dropdown arrow → select "Single Value (list)" or "Multiple values (list)".

Date Range Filter

Step 1: Drag [timestamp] from daily_kpis.csv to the Filters shelf.

Step 2: In the filter dialog, select "Range of Dates".

Step 3: Set default range: last 30 days.

Step 4: Right-click → Show Filter → Apply to all sheets using Daily KPIs.

10.2 Dashboard Actions (Interactions)

Dashboard Actions enable interactivity like clicking a chart to filter others. Set them up via Dashboard → Actions.

Filter Action 1: Click Geo Table to Filter Charts

Step 1: Go to Dashboard menu → Actions.

Step 2: Click "Add Action" → "Filter".

Step 3: Name: "Geo Filter".

Step 4: Source Sheets: select "Geo Table".

Step 5: Run action on: "Select".

Step 6: Target Sheets: select all other worksheets on the dashboard.

Step 7: Clearing the selection: "Show all values".

Step 8: Click OK.

Filter Action 2: Click Category Donut to Filter Trend Chart

Step 1: Add another Filter Action.

Step 2: Source: "Category Donut".

Step 3: Target: "Category Trends", "Engagement Trend", "Conversation Funnel".

Step 4: Fields: [prompt_category] → [prompt_category].

Highlight Action: Hover to Highlight Related Data

Step 1: Add Action → Highlight.

Step 2: Source: Any worksheet. Run on: "Hover".

Step 3: Target: select correlated worksheets.

Step 4: Fields: select the shared dimension (e.g., [language] or [prompt_category]).

10.3 Navigation Between Dashboards (Tabs)

Create navigation buttons that link between the three dashboards:

Step 1: Go to Dashboard 1 (Executive Summary).

Step 2: From the Objects panel, drag "Navigation" to the Tab Bar horizontal container.

Step 3: In the dialog, set "Navigate to": Dashboard 2 (Operational Intelligence).

Step 4: Button style: "Text only" — label: "⚙ OPERATIONAL INTELLIGENCE".

Step 5: Font: Segoe UI, 12px, Bold, uppercase. Color: #9A9FB0 (inactive) — Tableau doesn't support conditional styling easily, so use a consistent style.

Step 6: Repeat: add a Navigation button to Dashboard 3 as well.

Step 7: On Dashboard 2 and 3, add navigation buttons back to Dashboard 1 and to each other.

11. Final Formatting & Publishing

11.1 Global Formatting Settings

Apply these formatting rules across all worksheets for consistency:

Default Font Settings

Step 1: Format menu → Workbook → click "All" (applies to entire workbook).

Step 2: Font: Segoe UI, size 10 (base). Bold for headings.

Step 3: For KPI values, you will override the font per worksheet to Georgia.

Remove Default Borders and Backgrounds

Step 1: For each worksheet, go to Format → Borders.

Step 2: Set all borders to None except Row Divider (set to 1px #E0E0E0).

Step 3: Remove sheet borders: Format → Border → Sheet Border → None.

Axis Formatting

Step 1: Right-click each axis → Format.

Step 2: Axis font: Segoe UI, 10px, color #9A9FB0 (muted gray).

Step 3: Axis titles: Segoe UI, 11px, bold, color #1A1A2E.

Step 4: Axis tick marks: set to None or minimal.

Tooltip Formatting

Step 1: Click the Tooltip shelf on any worksheet.

Step 2: Clear default tooltip. Write a custom tooltip with relevant fields.

Step 3: Example for Geo Table: "Country: [country] | Sessions: [SUM(conversation_count)] | Toxicity: [AVG(toxicity_rate)]%"

Step 4: Font: Segoe UI, 11px. Background: #FFFFFF.

11.2 Dashboard-Level Formatting

Step 1: Click on the dashboard canvas (not a worksheet).

Step 2: Format menu → Dashboard.

Step 3: Title: set visibility to "Hidden" (title is embedded in the topbar text object).

Step 4: Padding: Inner padding → 20px on all sides.

Step 5: Shading: Background → #F5F5F5 (matches the HTML page background).

11.3 Adding Chart Panel Headers

Each chart panel in the HTML has a panel header bar (gray background, title + subtitle). Recreate this in Tableau:

Step 1: In each worksheet, double-click the title area to edit it.

Step 2: Use the title for the chart name (e.g., "WEEKLY ENGAGEMENT & DROP-OFF TREND").

Step 3: Format title: Segoe UI, 12px, Bold, UPPERCASE, #1A1A2E.

Step 4: Subtitle: Add a caption via Worksheet → Show Caption. Edit the caption text.

Step 5: Caption formatting: Segoe UI, 11px, normal, #9A9FB0.

11.4 Saving and Exporting

Save as Packaged Workbook

Step 1: File menu → Save As.

Step 2: Change file type dropdown to "Tableau Packaged Workbook (.twbx)".

Step 3: Name the file: "WildChat_Analytics_Dashboard.twbx".

Step 4: Click Save. This bundles the workbook WITH all data sources so anyone can open it without needing the original CSV files.

📌 **NOTE:** A .twbx file includes both the workbook definition and the data extracts. Share this single file to distribute the dashboard.

Publish to Tableau Public (Optional)

Step 1: Server menu → Tableau Public → Save to Tableau Public.

Step 2: Sign in with your Tableau Public account (free).

Step 3: Give your viz a name and click Save.

Step 4: Tableau Public will open in your browser with a shareable link.

Export as PDF or Image

Step 1: File menu → Print to PDF.

Step 2: In the dialog, select "Entire Workbook" to export all three dashboards.

Step 3: Paper size: Legal (14×8.5 landscape) for best fit.

Step 4: Click OK to save.

11.5 Quick Reference Summary — Dashboard Build Checklist

#	Task	Status
1	Connected all 6 CSV data sources	☐
2	Set geographic roles for country fields	☐
3	Created all 14 calculated fields	☐
4	Built all 5 Executive Summary worksheets	☐
5	Built all 12 Operational Intelligence worksheets	☐
6	Built all 9 Safety & Trust worksheets	☐
7	Created Dashboard 1 layout (Executive Summary)	☐
8	Created Dashboard 2 layout (Operational Intelligence)	☐
9	Created Dashboard 3 layout (Safety & Trust)	☐
10	Added filter controls and linked to all sheets	☐
11	Set up Dashboard Actions (filter/highlight)	☐
12	Added navigation buttons between dashboards	☐
13	Applied color palette (all HEX codes verified)	☐
14	Formatted fonts (Segoe UI + Georgia for KPIs)	☐
15	Set dashboard background to #F5F5F5	☐
16	Set panel backgrounds to #FFFFFF with borders	☐
17	Customized all tooltips	☐
18	Saved as .twbx packaged workbook	☐

End of WildChat Analytics Platform — Tableau Dashboard Recreation Guide

Group G-6 | For support, reference the HTML dashboard source file alongside this guide.